

Nils Froleyks

KU Leuven, Belgium

nfroleyks@gmail.com — [+31 616112049](tel:+31616112049) — [Google Scholar](#) — [GitHub](#)

RESEARCH INTERESTS

Certification, Artificial Intelligence, Automated Reasoning, Hardware Verification, and Planning.

Using formal proofs as a principled way to bridge Artificial Intelligence and Automated Reasoning at scale.

ACADEMIC EXPERIENCE

Postdoctoral Researcher

October 2025 – Present

KU Leuven, Belgium, Declarative Languages and Artificial Intelligence, Prof. Bart Bogaerts

Research on machine-checkable certification and automated reasoning at the intersection of formal methods and AI.

University Assistant

March 2020 – August 2025

Johannes Kepler University Linz, Austria, Formal Models and Verification, Prof. Armin Biere

and Symbolic Artificial Intelligence, Prof. Martina Seidl

Research in SAT solving and model checking; lecturer, teaching assistant, and bachelor's-thesis supervisor.

EDUCATION

Johannes Kepler University Linz, Austria

March 2020 – August 2025

Ph.D. in Computer Science with Distinction

Thesis: Deep Integration of SAT Solving and Model Checking

Supervisors: Prof. Armin Biere, Prof. Martina Seidl

Karlsruhe Institute of Technology, Germany

2018 – 2020

M.Sc. in Computer Science

Best Thesis Award

Thesis: PASAR—Planning as Satisfiability with Abstraction Refinement

Supervisors: Prof. Peter Sanders

Karlsruhe Institute of Technology, Germany

2013 – 2017

B.Sc. in Computer Science

Thesis: Using an Algorithm Portfolio to Solve Sokoban

Supervisors: Prof. Peter Sanders

INDUSTRIAL COLLABORATION

Intel Corporation

Nov 2023 – Present

Continued project on integrating hardware model checking certification into industrial workflows.

Woodpecker Technologies, Singapore

2022 – 2023

External consultant. Pre-silicon verification using symbolic formal methods.

CAREER BREAKS

Parental Leave

February – November 2022

AWARDS

CAV Distinguished Paper Award

2026

38th International Conference on Computer Aided Verification

for *Liveness Proofs for Hardware Model Checking*

Fahiem Bacchus Award runner-up

2026

Awarded by the SAT Association for outstanding PhD thesis in the field of satisfiability

Heinz Zemanek Award runner-up

2026

Austrian Computer Society (OCG) award for outstanding PhD theses in computer science

CAV Distinguished Paper Award

2025

37th International Conference on Computer Aided Verification

for *Introducing Certificates to the Hardware Model Checking Competition*

Third Place in Combinatorial Reconfiguration Challenge

2022

CoRe Challenge 2022, Virtual

Best Master's Thesis

2020

Karlsruhe Institute of Technology, Germany

First Prize (Partial Gold Medal)

2019

The Sparkle Planning Challenge, ICAPS, Berkeley, USA

COMMUNITY AND PROJECT LEADERSHIP

Hardware Model Checking Competition: organizer: 2020 – 2027;
lead the introduction of mandatory certification used since 2024.

SAT Competition: organizer: 2020, 2021; co-author of the 2020 journal report in *Artificial Intelligence*.

Reviewer: SAT (2022, 2024, 2025); SoCS (2022); TACAS (2023, 2025); SEFM (2024); iFM (2024); Journal of Artificial Intelligence (2025–2026); ISEDA (2026); IJCAI (2026); Journal of Combinatorial Theory, Series A (2026); TACAS Artifact Evaluation (2026).

TEACHING & SUPERVISION

Lecturer / guest lecturer: Model Checking and Debugging, JKU (2024); SAT Solving, KUL (2026); Formal Methods for AI, Leiden University (2026).

Master's-thesis committees: Integrating Graph Generation Algorithms into Large Language Models, KUL 2026; Diverse Explanations of Unsatisfiability, KUL 2026; Longest Cycles with few Chords in 3-Connected Graphs, KUL 2026.

Bachelor's-thesis supervision: Certifying Constraints from Hardware Designs, Leiden University (2026); Ternary Simulation for IC3 Model Checking, JKU (2024).

Teaching assistant, JKU (2020–2024): hands-on bachelor's and master's courses in Model Checking, Formal Models, Logic, SAT Solving, Practical Work in AI, and Computational Logics for AI.

Teaching assistant, KIT: Theoretical Foundations of Informatics (2015–2019) and Algorithms for Planar Graphs (2018).

SELECTED OPEN-SOURCE SOFTWARE

Certifaiger competition-adopted bit-level certificate checker
CaDiCaL award-winning incremental SAT solver
Voirraig certifying model checker
Cerbtora word-level certificate checker
CadiBack high-performance backbone extractor

RESEARCH VISITS

Prof. Mingshuai Chen, Zhejiang University	August 2026
Dr. Adrián Rebola Pardo, TU Vienna	May 2026
Prof. Matti Järvisalo, University of Helsinki	November 2025
Prof. Bart Bogaerts, KU Leuven	July 2025
Prof. Joost-Pieter Katoen, RWTH Aachen	June 2024

SKILLS

Programming Languages: C++, Python
Languages: German (native), English (fluent)

CONTACTS FOR REFERENCES

Prof. Bart Bogaerts, KU Leuven, Belgium. bart.bogaerts@kuleuven.be

Prof. Armin Biere, University of Freiburg, Germany. biere@cs.uni-freiburg.de

Prof. Martina Seidl, Johannes Kepler University Linz, Austria. martina.seidl@jku.at

PUBLICATIONS

- [1] **Nils Froleys**, Emily Yu, Bart Bogaerts, Armin Biere, and Keijo Heljanko. *Liveness Proofs for Hardware Model Checking*. International Conference on Computer Aided Verification (CAV), 2026.
Distinguished Paper Award
- [2] Florian Pollitt, Mathias Fleury, Katalin Fazekas, **Nils Froleys**, André Schidler, Dominik Schreiber, and Armin Biere. *CaDiCaL 3.0: Proofs, Algorithms, Implied Literals, and Robust Scheduling*. Theory and Applications of Satisfiability Testing (SAT), 2026.
- [3] **Nils Froleys** and Emily Yu. *Hardware Model Checking Certification with Certifaiger and Cerbtora*. International Joint Conference on Automated Reasoning (IJCAR), 2026.
- [4] Armin Biere, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *Clausal Equivalence Checking*. Journal manuscript under review.
- [5] **Nils Froleys**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Constraints in Hardware Model Checking*. International Symposium on Formal Methods (FM), 2026.
- [6] **Nils Froleys**, Emily Yu, Mathias Preiner, Armin Biere, and Keijo Heljanko. *Introducing Certificates to the Hardware Model Checking Competition*. International Conference on Computer Aided Verification (CAV), 2025.
Distinguished Paper Award
- [7] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *CaDiCaL 2.0*. International Conference on Computer Aided Verification (CAV), 2024.
- [8] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleys**. *Clausal Equivalence Sweeping*. Formal Methods in Computer-Aided Design (FMCAD), 2024.
- [9] Armin Biere, **Nils Froleys**, and Mathias Preiner. *Hardware Model Checking Competition 2024*. Formal Methods in Computer-Aided Design (FMCAD), 2024.
- [10] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleys**. *Clausal Congruence Closure*. Theory and Applications of Satisfiability Testing (SAT), 2024.
- [11] **Nils Froleys**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Phase Abstraction*. International Joint Conference on Automated Reasoning (IJCAR), 2024.
- [12] **Nils Froleys**, Emily Yu, and Armin Biere. *Ternary Simulation as Abstract Interpretation*. Methoden und Beschreibungssprachen zur Modellierung und Verifikation von Schaltungen und Systemen (MBMV), 2024.
- [13] Armin Biere, **Nils Froleys**, and Wenxi Wang. *CadiBack: Extracting Backbones with CaDiCaL*. Theory and Applications of Satisfiability Testing (SAT), 2023.
- [14] Armin Biere, Mathias Fleury, **Nils Froleys**, and Marijn JH Heule. *The SAT Museum*. Pragmatics of SAT, 2023.
- [15] **Nils Froleys**, Emily Yu, and Armin Biere. *BIG Backbones*. Formal Methods in Computer-Aided Design (FMCAD), 2023.
- [16] Emily Yu, **Nils Froleys**, Armin Biere, and Keijo Heljanko. *Towards Compositional Hardware Model Checking Certification*. Formal Methods in Computer-Aided Design (FMCAD), 2023.
- [17] Emily Yu, **Nils Froleys**, Armin Biere, and Keijo Heljanko. *Stratified Certification for K-Induction*. Formal Methods in Computer-Aided Design (FMCAD), 2022.
- [18] **Nils Froleys** and Armin Biere. *Single Clause Assumption without Activation Literals to Speed-up IC3*. Formal Methods in Computer-Aided Design (FMCAD), 2021.
- [19] Tomáš Balyo and **Nils Froleys**. *AI Assisted Design of Sokoban Puzzles Using Automated Planning*. International Conference on ArtsIT, Interactivity and Game Creation, 2021.
- [20] **Nils Froleys**, Marijn Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *SAT Competition 2020*. Artificial Intelligence, 2021.
- [21] **Nils Froleys**, Tomáš Balyo, and Dominik Schreiber. *PASAR—Planning as Satisfiability with Abstraction Refinement*. International Symposium on Combinatorial Search (SoCS), 2019.
- [22] **Nils Froleys** and Tomáš Balyo. *Using an Algorithm Portfolio to Solve Sokoban*. International Symposium on Combinatorial Search (SoCS), 2017.

TECHNICAL REPORTS AND COMPETITION PROCEEDINGS

- [1] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, Florian Pollitt, and André Schidler. *CaDiCaL, IsaSAT and Kissat Entering the SAT Competition 2025*. SAT Competition 2025.
- [2] **Nils Froleys**, Emily Yu, and Armin Biere. *Challenging Certificates from Model Checking*. SAT Competition 2025.
- [3] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *Hardware Equivalence Checking Problems Submitted to the SAT Competition 2024*. SAT Competition 2024.
- [4] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *CaDiCaL, Gimsatul, IsaSAT and Kissat Entering the SAT Competition 2024*. SAT Competition 2024.
- [5] **Nils Froleys**, Emily Yu, and Armin Biere. *ReconfAIGERation entering Core Challenge 2022*. CoRe Challenge 2022.
- [6] **Nils Froleys**, Emily Yu, and Armin Biere. *Unique Reconfiguration Sequence*. SAT Competition 2022.
- [7] Emily Yu, **Nils Froleys**, Armin Biere, and Mathias Fleury. *Hardware Model Checking Certificates*. SAT Competition 2022.
- [8] Tomáš Balyo, **Nils Froleys**, Marijn JH Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *Proceedings of SAT Competition 2020: Solver and Benchmark Descriptions*. 2020.
- [9] **Nils Froleys**, Tomáš Balyo, and Dominik Schreiber. *PASAR Entering the Sparkle Planning Challenge 2019*. Sparkle Planning Challenge 2019.